LAB EXPERIMENTS

TOOL: R Programming **RegisterNo:192424189**

Name: Dhanusha D **Date:20-07-2026**

COURSE: Data Handling and Visualization **Code: DSA0604**

SET-6

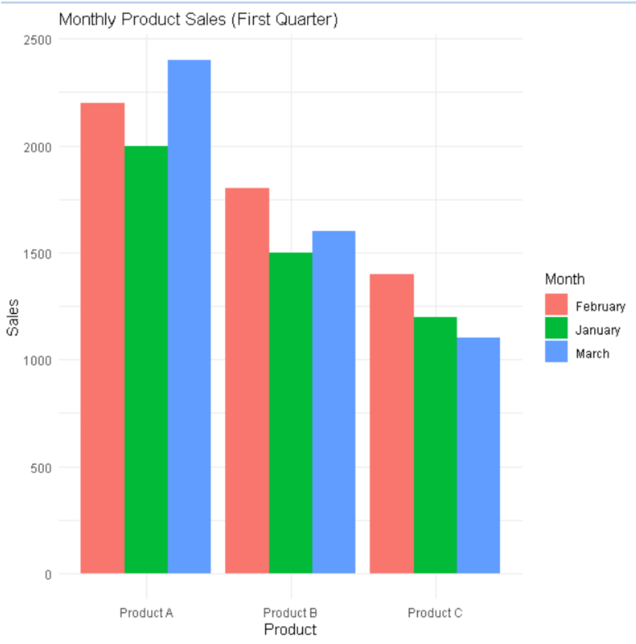
Question 1: Grouped Bar Chart

R PROGRAM:

```
library(ggplot2)
library(tidyr)
sales <- data.frame(
  Product = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)
sales_long <- pivot_longer(
  sales,
  cols = January:March,
  names_to = "Month",
  values_to = "Sales"
)
ggplot(sales_long,
       aes(x = Product,
           y = Sales,
           fill = Month)) +
  geom_bar(stat = "identity",
           position = "dodge") +
  labs(
    title = "Monthly Product Sales (First Quarter)",
    x = "Product",
```

```
  y = "Sales"
 ) +
 theme_minimal()
```

OUTPUT:

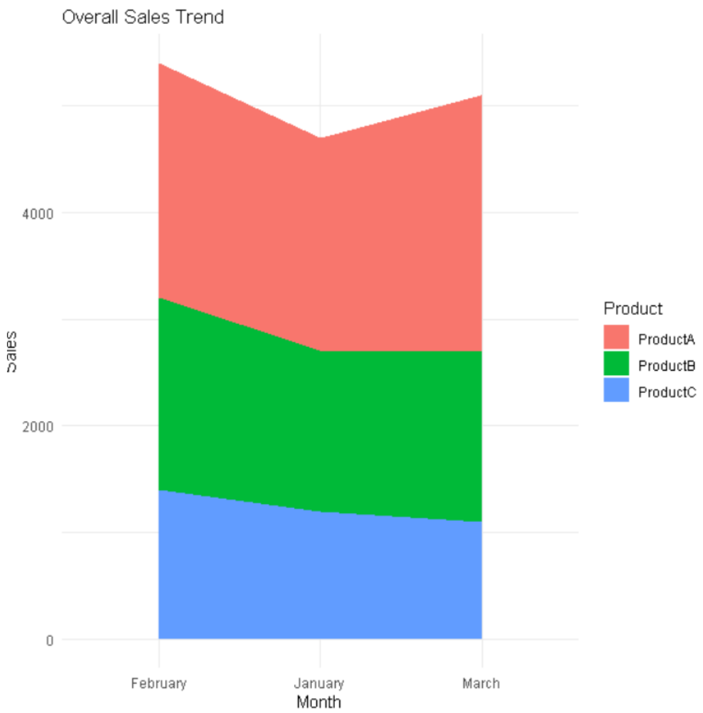


Question 2: Stacked Area Chart

R PROGRAM:

```
library(ggplot2)
library(tidyr)
sales <- data.frame(
 Month = c("January","February","March"),
 ProductA = c(2000,2200,2400),
 ProductB = c(1500,1800,1600),
 ProductC = c(1200,1400,1100)
)
sales_long <- pivot_longer(
 sales,
 cols = ProductA:ProductC,
 names_to = "Product",
 values_to = "Sales"
```

```
)
ggplot(sales_long,
       aes(x = Month,
           y = Sales,
           fill = Product,
           group = Product)) +
  geom_area() +
  labs(
    title = "Overall Sales Trend",
    x = "Month",
    y = "Sales"
  ) +
  theme_minimal()
```



Question 3: Monthly Sales Table

R PROGRAM:

```
sales <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),
```

```
  JanuarySales = c(2000,1500,1200),
  FebruarySales = c(2200,1800,1400),
  MarchSales = c(2400,1600,1100)
)
print(sales)
```

```
print(sales)
ProductID ProductName JanuarySales FebruarySales MarchSales
        1   Product A         2000          2200       2400
        2   Product B         1500          1800       1600
        3   Product C         1200          1400       1100
# Question 3: Monthly Sales Table

sales <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),
  JanuarySales = c(2000,1500,1200),
  FebruarySales = c(2200,1800,1400),
  MarchSales = c(2400,1600,1100)
)

print(sales)
ProductID ProductName JanuarySales FebruarySales MarchSales
        1   Product A         2000          2200       2400
        2   Product B         1500          1800       1600
        3   Product C         1200          1400       1100
```

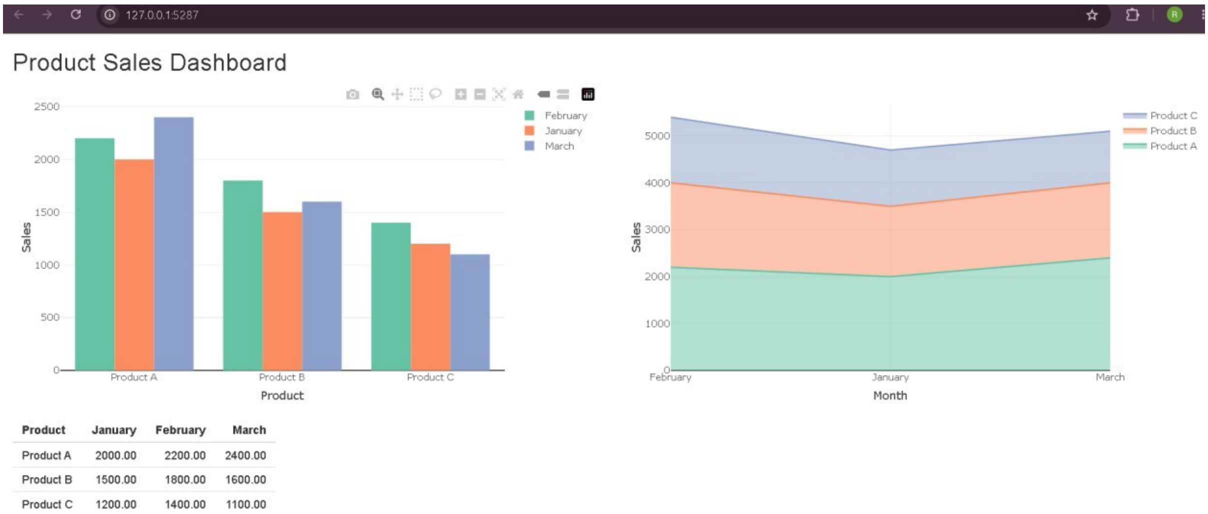
Question 4: Interactive Dashboard (Shiny + Plotly)

R PROGRAM:

```
library(shiny)
library(plotly)
library(tidyr)

sales <- data.frame(
  Product = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)

sales_long <- pivot_longer(
```

```
  sales,
  cols = January:March,
  names_to = "Month",
  values_to = "Sales"
)

ui <- fluidPage(

    titlePanel("Product Sales Dashboard"),

    fluidRow(

        column(
            6,
            plotlyOutput("barPlot")
        ),

        column(
            6,
            plotlyOutput("areaPlot")
        )

    ),

    br(),

    tableOutput("salesTable")

)
```

```
server <- function(input, output){
output$barPlot <- renderPlotly({
plot_ly(
      sales_long,
      x = ~Product,
      y = ~Sales,
      color = ~Month,
      type = "bar"
    ) %>%
      layout(barmode = "group")

  }) output$areaPlot <- renderPlotly({
 plot_ly(
      sales_long,
      x = ~Month,
      y = ~Sales,
      color = ~Product,
      type = "scatter",
      mode = "lines",
      stackgroup = "one"
    )
  })
  output$salesTable <- renderTable({
    sales
  })
}
shinyApp(ui, server)
```



Product	January	February	March
Product A	2000.00	2200.00	2400.00
Product B	1500.00	1800.00	1600.00
Product C	1200.00	1400.00	1100.00

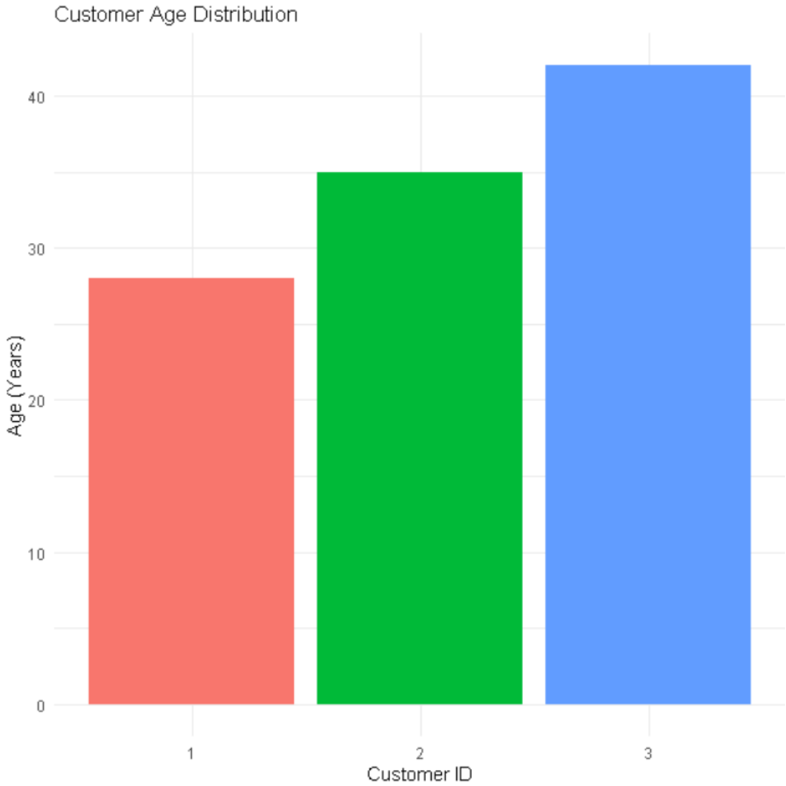
SET-7

Question 1: Bar Chart - Customer Age Distribution

R PROGRAM:

```
library(ggplot2)
customers <- data.frame(
CustomerID = c(1,2,3),
Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
ggplot(customers,
       aes(x = factor(CustomerID),
           y = Age,
           fill = factor(CustomerID))) +
  geom_bar(stat = "identity") +
  labs(
    title = "Customer Age Distribution",
    x = "Customer ID",
    y = "Age (Years)"
  ) +
```

```
  theme_minimal() +
  theme(legend.position = "none")
```

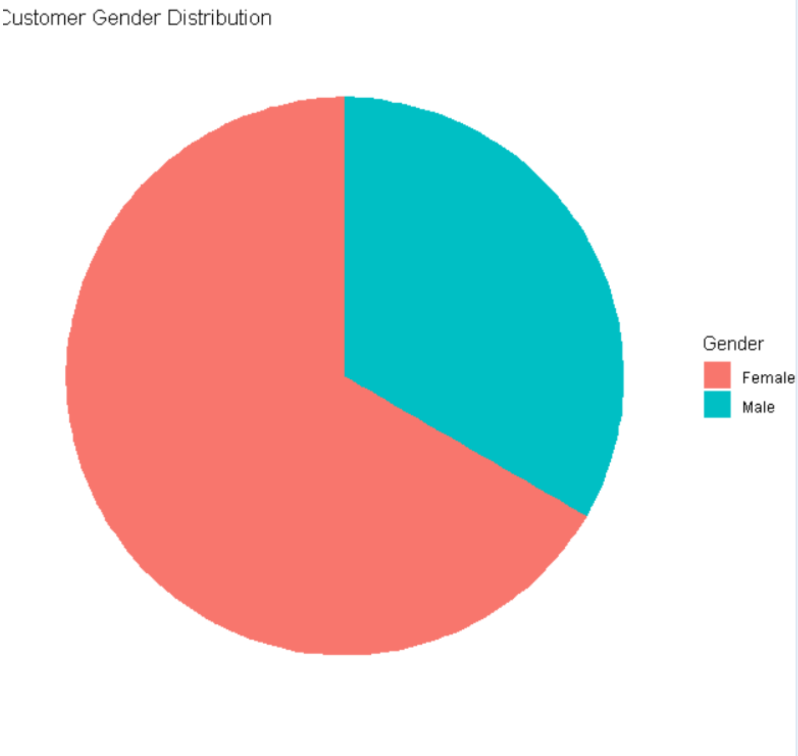


Question 2: Pie Chart - Gender Distribution

R PROGRAM

```
library(ggplot2)
customers <- data.frame(
  Gender = c("Female","Male","Female")
)
gender_count <- as.data.frame(table(customers$Gender))
colnames(gender_count) <- c("Gender","Count")
ggplot(gender_count,
       aes(x = "",
           y = Count,
           fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(
```

```
    title = "Customer Gender Distribution"
  ) +
  theme_void()
```



Question 3: Customer Demographics Table

R PROGRAM:

```
customers <- data.frame(
  CustomerID  =  c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
print(customers)
```

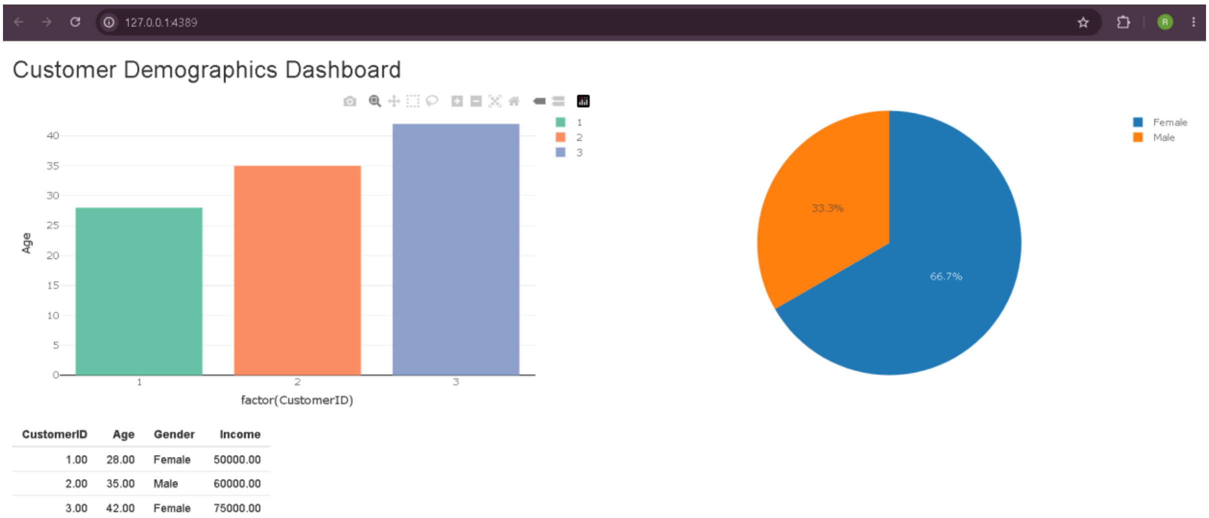
```
+ )
>
> print(customers)
  CustomerID Age Gender Income
1          1  28 Female  50000
2          2  35   Male  60000
3          3  42 Female  75000
>
```

Question 4: Interactive Dashboard (Shiny + Plotly)

R PROGRAM:

```
library(shiny)

library(plotly)

customers <- data.frame(

 CustomerID = c(1,2,3),

 Age = c(28,35,42),

 Gender = c("Female","Male","Female"),

 Income = c(50000,60000,75000)

)

gender_count <- as.data.frame(table(customers$Gender))

colnames(gender_count) <- c("Gender","Count")

\ui <- fluidPage(

 titlePanel("Customer Demographics Dashboard"),

 fluidRow(

 column(

         6,

         plotlyOutput("barPlot")

       ),


       column(

         6,

         plotlyOutput("piePlot")

       )
```

```
    ),

    br(),

    tableOutput("customerTable")

)server <- function(input, output){
output$barPlot <- renderPlotly({
plot_ly(
            customers,
            x = ~factor(CustomerID),
            y = ~Age,
            type = "bar",
            color = ~factor(CustomerID)
        )
  })output$piePlot <- renderPlotly({
        plot_ly(
            gender_count,
            labels = ~Gender,
            values = ~Count,
            type = "pie"
        )
    })

    output$customerTable <- renderTable({
customers
    })
}
shinyApp(ui, server)
```



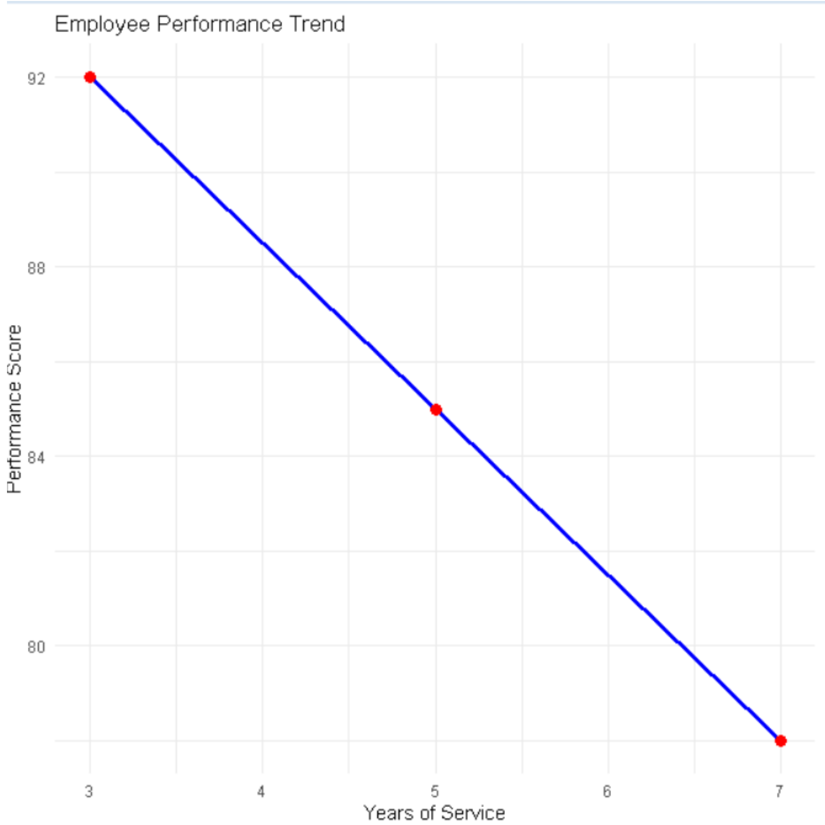
SET-8

Question 1: Line Chart - Employee Performance Trend

R PROGRAM:

```
library(ggplot2)
employee <- data.frame(
EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsOfService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
ggplot(employee,
       aes(x = YearsOfService,
           y = PerformanceScore,
           group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Employee Performance Trend",
    x = "Years of Service",
    y = "Performance Score"
```

```
 ) +
```

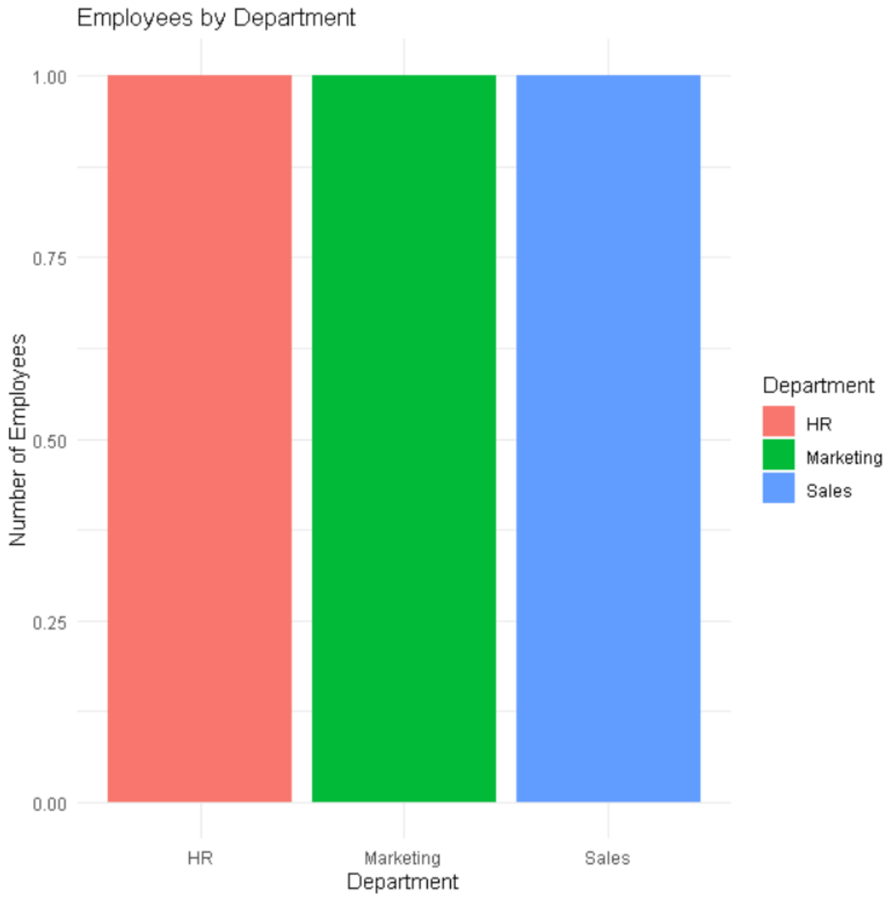


Question 2: Bar Chart - Employees by Department

R PROGRAM:

```
library(ggplot2)
employee <- data.frame(
  Department = c("Sales","HR","Marketing")
)
dept_count <- as.data.frame(table(employee$Department))
colnames(dept_count) <- c("Department","Count")
ggplot(dept_count,
       aes(x = Department,
           y = Count,
           fill = Department)) +
  geom_bar(stat = "identity") +
```

```
labs(
  title = "Employees by Department",
  x = "Department",
  y = "Number of Employees"
) +
theme_minimal()
```



Question 3: Employee Performance Table

R PROGRAM:

```
employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsOfService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
print(employee)
```

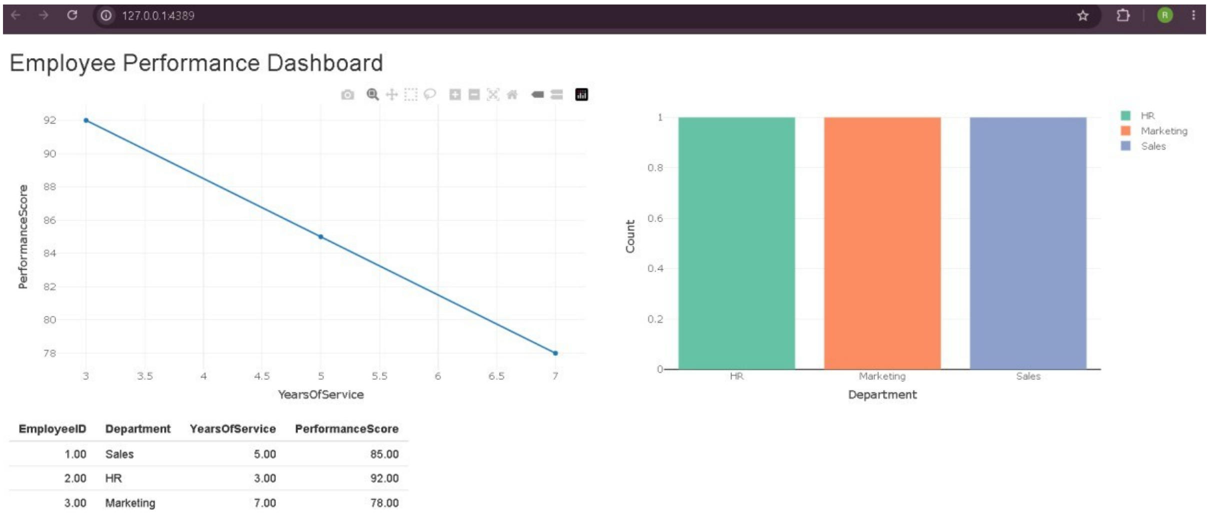
```
> employee <- data.frame(
+   EmployeeID = c(1,2,3),
+   Department = c("Sales","HR","Marketing"),
+   YearsOfService = c(5,3,7),
+   PerformanceScore = c(85,92,78)
+ )
>
> print(employee)
  EmployeeID Department YearsOfService PerformanceScore
1          1      Sales              5               85
2          2         HR              3               92
3          3  Marketing              7               78
> |
```

Question 4: Interactive Dashboard (Shiny + Plotly)

R PROGRAM:

```
library(shiny)
library(plotly)
employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsOfService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
dept_count <- as.data.frame(table(employee$Department))
colnames(dept_count) <- c("Department","Count")
ui <- fluidPage(
titlePanel("Employee Performance Dashboard"),
fluidRow(
column(
      6,
      plotlyOutput("linePlot")
    ),
column(
      6,
      plotlyOutput("barPlot")
    )
```

```
  ),
 br(),
tableOutput("employeeTable"))
server <- function(input, output){
output$linePlot <- renderPlotly({
plot_ly(
      employee,
      x = ~YearsOfService,
      y = ~PerformanceScore,
      type = "scatter",
      mode = "lines+markers"
    )
  })
output$barPlot <- renderPlotly({
  plot_ly(
      dept_count,
      x = ~Department,
      y = ~Count,
      type = "bar",
      color = ~Department
    )
  })
 output$employeeTable <- renderTable({
 employee
  })
}
shinyApp(ui, server)
```



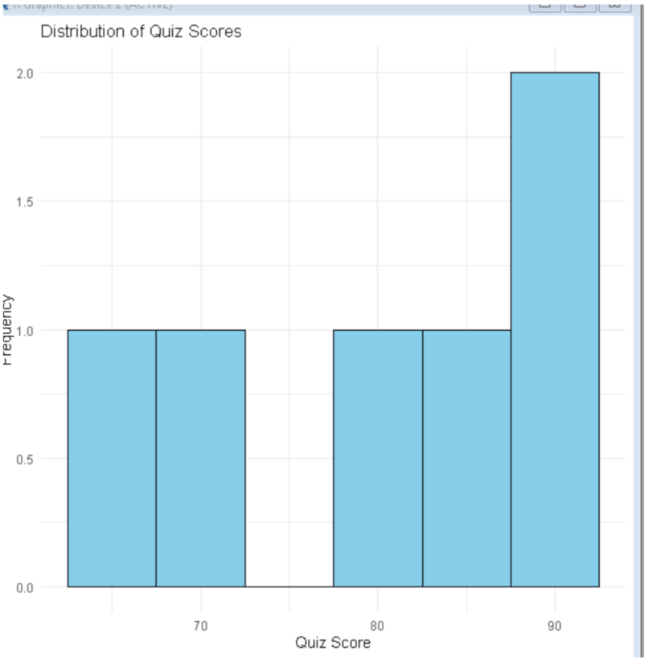
SET -9

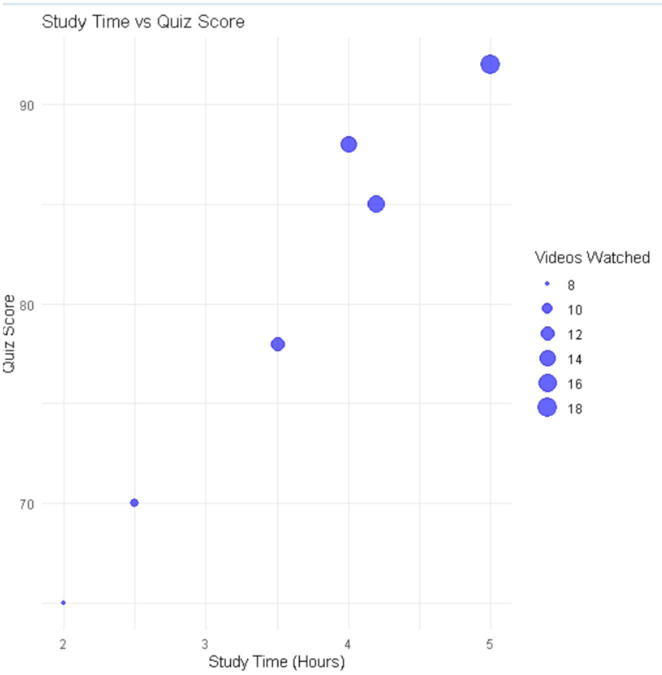
Question 1: Import Dataset, Histogram & Boxplot

R PROGRAM:

```
library(ggplot2)
data <- read.csv("Online_Learning_Activity.csv")
ggplot(data, aes(x = Quiz_Score)) +
geom_histogram(binwidth = 5,
               fill = "skyblue",
               color = "black") +
  labs(
    title = "Distribution of Quiz Scores",
    x = "Quiz Score",
    y = "Frequency"
  ) +
  theme_minimal()
ggplot(data,
       aes(x = Course,
           y = Quiz_Score,
           fill = Course)) +
  geom_boxplot() +
```

```
labs(
  title = "Quiz Score by Course",
  x = "Course",
  y = "Quiz Score"
) +
theme_minimal()
```

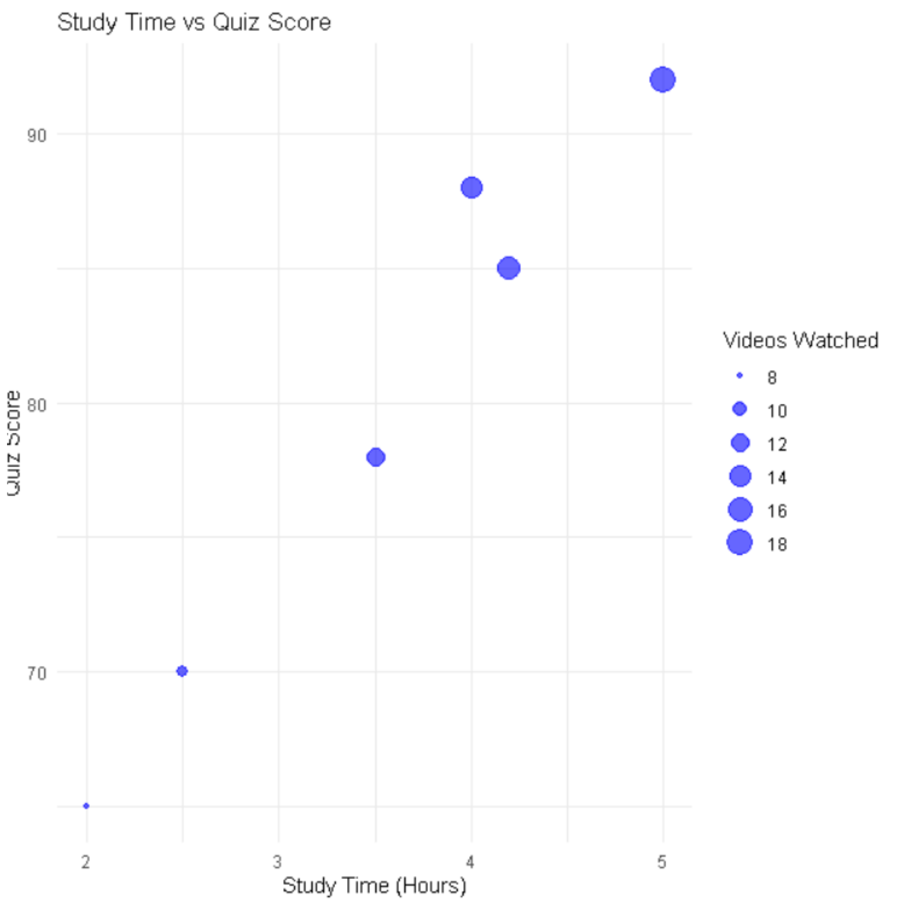




Question 2: Bubble Scatter Plot

R PROGRAM:

```
library(ggplot2)

data <- read.csv("Online_Learning_Activity.csv")
ggplot(data,
       aes(x = Study_Time,
           y = Quiz_Score,
           size = Videos_Watched)) +
geom_point(alpha = 0.6,
             color = "blue") +
  labs(
    title = "Study Time vs Quiz Score",
    x = "Study Time (Hours)",
    y = "Quiz Score",
    size = "Videos Watched"
  ) +
theme_minimal()
```



Question 3: Monthly Average Quiz Score with Moving Average

R PROGRAM:

```
library(ggplot2)
library(dplyr)
library(zoo)
data <- read.csv("Online_Learning_Activity.csv")
data$Login_Date <- as.Date(data$Login_Date)
data$Month <- format(data$Login_Date, "%Y-%m")
monthly <- data %>%
  group_by(Month) %>%
summarise(Average_Quiz = mean(Quiz_Score))
monthly$Moving_Average <- rollmean(
monthly$Average_Quiz,
  k = 2,
  fill = NA,
```

```
  align = "right"
)
ggplot(monthly,
       aes(x = Month,
           y = Average_Quiz,
           group = 1)) +
 geom_line(color = "blue",
           linewidth = 1.2) +

  geom_point(size = 3,
             color = "red") +

  geom_line(aes(y = Moving_Average),
           color = "green",
           linewidth = 1.2,
           linetype = "dashed") +

  labs(
    title = "Monthly Average Quiz Score",
    x = "Month",
    y = "Average Quiz Score"
  ) +

  theme_minimal()
```

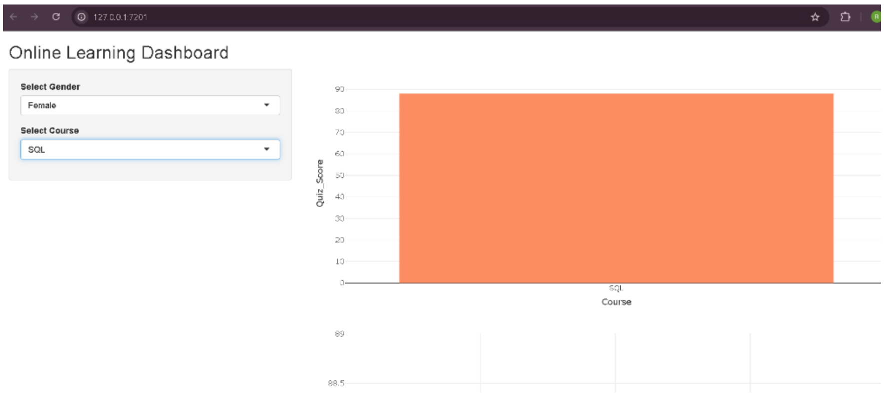
Question 4: Interactive Dashboard (Shiny + Plotly)

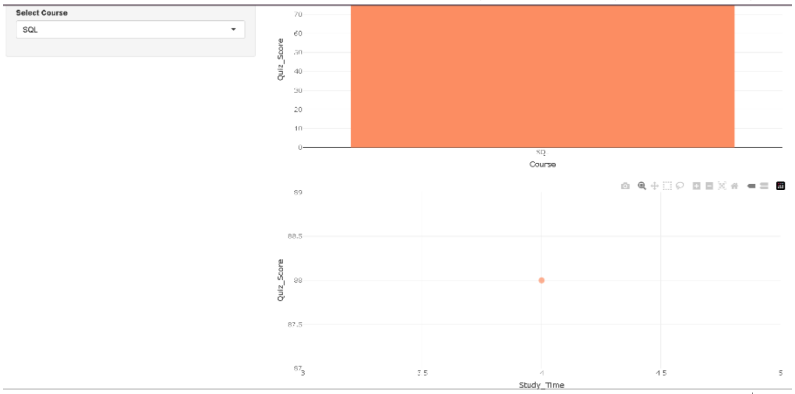
R PROGRAM:

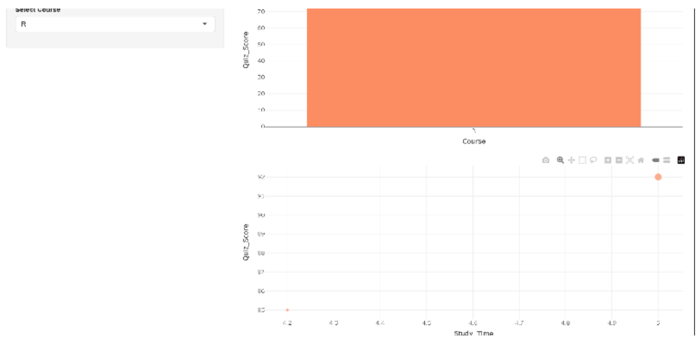
```
library(shiny)

library(plotly)

data <- read.csv("Online_Learning_Activity.csv")

ui <- fluidPage(

titlePanel("Online Learning Dashboard"),

sidebarLayout(

sidebarPanel(

 selectInput(

        "gender",

        "Select Gender",

        choices = c("All",

                 unique(data$Gender))

      ),


      selectInput(

        "course",

        "Select Course",

        choices = c("All",
```

```
                        unique(data$Course))
          )

        ),
 mainPanel(
plotlyOutput("barPlot"),
br(),plotlyOutput("scatterPlot")

        )

    )

)

server <- function(input, output){

    filtered <- reactive({

        df <- data

        if(input$gender != "All")
            df <- df[df$Gender == input$gender,]

        if(input$course != "All")
            df <- df[df$Course == input$course,]

        df

    })
```

```
  output$barPlot <- renderPlotly({

    avg <- aggregate(
      Quiz_Score ~ Course,
      data = filtered(),
      mean
    )

    plot_ly(
      avg,
      x = ~Course,
      y = ~Quiz_Score,
      type = "bar",
      color = ~Course
    )

  })output$scatterPlot <- renderPlotly({   plot_ly(
      filtered(),
      x = ~Study_Time,
      y = ~Quiz_Score,
      type = "scatter",
      mode = "markers",
      size = ~Videos_Watched,
      color = ~Course
    )
  })
}
shinyApp(ui, server)
```





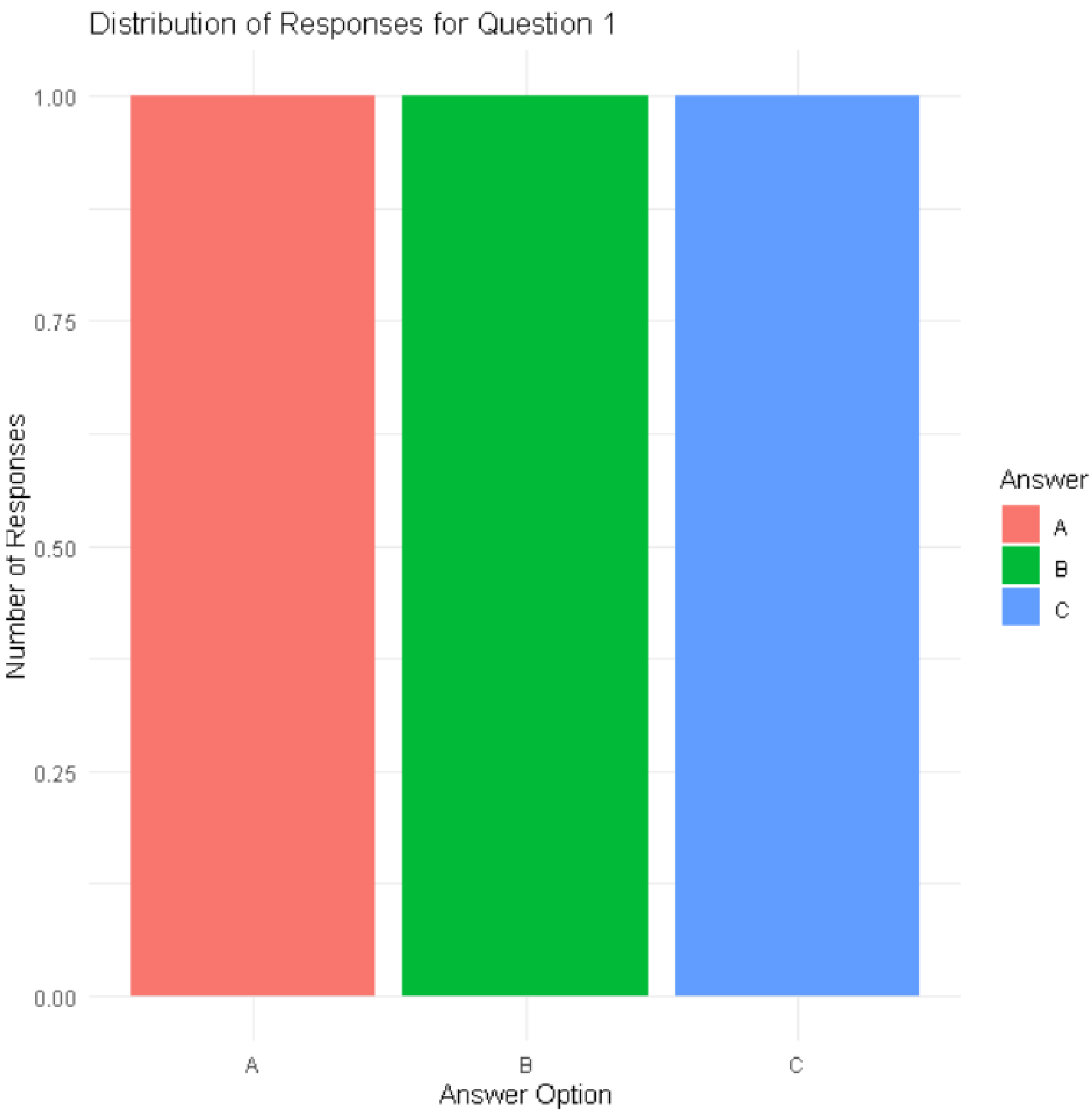


SET -10

Question 1: Grouped Bar Chart - Question 1 Responses

```
library(ggplot2)

survey <- data.frame(

SurveyID = c(1,2,3),
```

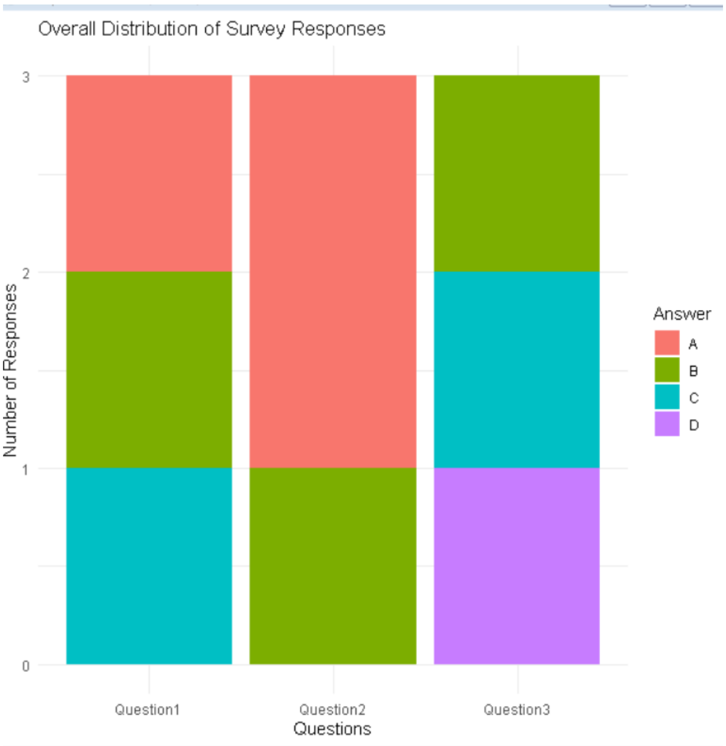
```
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
q1_count <- as.data.frame(table(survey$Question1))
colnames(q1_count) <- c("Answer","Count")
ggplot(q1_count,
       aes(x = Answer,
           y = Count,
           fill = Answer)) +
  geom_bar(stat = "identity",
           position = "dodge") +
  labs(
    title = "Distribution of Responses for Question 1",
    x = "Answer Option",
    y = "Number of Responses"
  ) +
  theme_minimal()
```



Question 2

```
library(ggplot2)
library(tidyr)
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
```

```
survey_long <- pivot_longer(
  survey,
  cols = Question1:Question3,
  names_to = "Question",
  values_to = "Answer"
)

ggplot(survey_long,
       aes(x = Question,
           fill = Answer)) +
  geom_bar() +
  labs(
    title = "Overall Distribution of Survey Responses",
    x = "Questions",
    y = "Number of Responses"
  ) +
  theme_minimal()
```



Question 3: Survey Response Table

R program:

```
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
print(survey)
```

```
+ )
>
> print(survey)
  SurveyID Question1 Question2 Question3
1        1         A         B         C
2        2         B         A         D
3        3         C         A         B
>
```

Question 4: Interactive Dashboard (Shiny + Plotly)

R PROGRAM:

```
library(shiny)
library(plotly)
library(tidyr)
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
q1_count <- as.data.frame(table(survey$Question1))
colnames(q1_count) <- c("Answer","Count")

survey_long <- pivot_longer(
  survey,
```

```
  cols = Question1:Question3,
  names_to = "Question",
  values_to = "Answer"
)
response_count <- as.data.frame(table(survey_long$Question,
                                      survey_long$Answer))
colnames(response_count) <- c("Question","Answer","Count")

ui <- fluidPage(
  titlePanel("Survey Responses Dashboard"),
  fluidRow(
 column(
      6,
      plotlyOutput("groupBar")
    ),
column(
      6,
      plotlyOutput("stackBar")
    )
  ),
 br(),
tableOutput("surveyTable")
)
server <- function(input, output){
output$groupBar <- renderPlotly({
plot_ly(
      q1_count,
      x = ~Answer,
      y = ~Count,
      type = "bar",
```

```
      color = ~Answer
    )
  })
  output$stackBar <- renderPlotly({
plot_ly(
      response_count,
      x = ~Question,
      y = ~Count,
      color = ~Answer,
      type = "bar"
    ) %>%
      layout(barmode = "stack")
  })
  output$surveyTable <- renderTable({
    survey
  })
}
shinyApp(ui, server)
```